

# Ayush Agarwal

+91 9122941107 | [ayush20039939@gmail.com](mailto:ayush20039939@gmail.com) | [linkedin.com/in/ayush20039939](https://linkedin.com/in/ayush20039939) | [github.com/Ayush99392003](https://github.com/Ayush99392003)

## SUMMARY

AI/ML engineer and CS undergraduate at VIT Bhopal with hands-on experience building production-grade LLM pipelines, RAG systems, and multi-agent architectures. Proficient in Python, PyTorch, Hugging Face Transformers, FastAPI, LangChain, and the Azure AI ecosystem. Seeking full-time or internship roles in AI/ML engineering, applied NLP, and LLM systems.

## EDUCATION

### VIT Bhopal University

*B.Tech in Computer Science and Engineering*

- CGPA: 8.52 / 10

Bhopal, India

Sep 2023 – Present

### D.A.V Public School

*Class XII*

- Percentage: 88.0%

Ranchi, India

May 2022

### D.A.V Nandraj Public School

*Class X*

- Percentage: 91.0%

Ranchi, India

May 2020

## EXPERIENCE

### AI/ML Intern

*Docu3C Technologies*

Aug 2025 – Present

*Remote*

- Developed and deployed AI/LLM-based applications for financial document intelligence using the Azure AI ecosystem including Azure ML, Cognitive Services, and GPU infrastructure.
- Built end-to-end LLM pipelines with structured outputs using Pydantic and schema validation via Instructor, reducing output parsing errors significantly.
- Designed multi-agent systems and tool-calling workflows using MCP (Model Context Protocol) to orchestrate complex document processing tasks.
- Built scalable FastAPI backends supporting real-time streaming (SSE/WebSockets) and async execution for high-throughput document workflows.
- Developed interactive UI applications using Streamlit and React for AI-driven validation and review systems.
- Maintained production-grade code quality using PEP8 standards, Ruff linting, and modular architecture; utilized uv for efficient dependency management.
- Conducted benchmarking, evaluation, and iterative optimization of AI systems using OpenTelemetry and Arize Phoenix for observability.

## PROJECTS

### Hybrid Legal RAG: Sparse–Dense–Graph Retrieval | *Python, FastAPI, FAISS, BM25, Docker* | [GitHub](#) 2024

- Engineered a production-grade retrieval-augmented generation system combining BM25 (FTS5), dense embeddings (MPNet + FAISS), and graph-based ranking (PageRank) over 26K+ indexed legal documents.
- Implemented Reciprocal Rank Fusion (RRF) to unify sparse, dense, and graph retrieval modalities, achieving **82.4% Top-1 accuracy** and **NDCG@10 of 0.89**, outperforming single-modality baselines.
- Built a FastAPI backend supporting semantic search, conversational querying, and structured summarization; containerized with Docker using memory-efficient optimizations.
- Conducted ablation studies to validate system robustness under high-noise conditions.

### Knowledge Graph Pipeline for Document Intelligence | *Python, spaCy, NetworkX, PyVis* | [GitHub](#) 2024

- Built an end-to-end knowledge graph extraction pipeline from unstructured documents using NLP and transformer-based LLMs with multi-stage processing: OCR, cleaning, chunking, entity-relation extraction, and validation.
- Designed deterministic ID mapping and cross-chunk entity resolution using fuzzy matching (RapidFuzz); exported structured graph data for downstream analytics.
- Visualized interactive knowledge graphs using NetworkX and PyVis for exploratory analysis.

### PolarBrief AI – Legal Argument Analyzer | *Python, LangChain, Streamlit, Tesseract OCR* | [GitHub](#) 2024

- Developed an AI-powered document analysis tool to extract, classify, and score arguments from unstructured PDFs using LangChain LLMs for summarization, heading generation, and structured scoring.
- Implemented argument polarity classification (Pro/Con/Neutral) with weighted ranking using TF-IDF combined with LLM outputs for improved precision.
- Designed a hybrid OCR + text extraction pipeline using Tesseract and PDF parsers; generated structured outputs in JSON/PDF with citation-aware summaries.

**WESAD Multimodal Emotion Recognition** | *Python, Scikit-learn, Pandas, NumPy* | [GitHub](#) 2024

- Implemented and benchmarked ML models (LDA, Random Forest, Decision Tree, AdaBoost, KNN) for emotion classification on multimodal physiological signals (ECG, EDA, EMG, BVP, ACC).
- Designed experiments evaluating sensor modality combinations (chest vs. wrist vs. combined); generated detailed reports identifying optimal modality configurations using F1-score and cross-validation.

TECHNICAL SKILLS

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**Languages:** Python, TypeScript  
**AI/ML:** Machine Learning, Deep Learning, NLP, LLMs, RAG, Prompt Engineering, Transformers  
**Frameworks & Libraries:** PyTorch, Hugging Face Transformers, LangChain, Scikit-learn, Pandas, NumPy, NLTK, spaCy, Pydantic, Instructor  
**Retrieval & Search:** FAISS, BM25 (FTS5), Knowledge Graphs, PageRank, RRF, Vector Databases, Embeddings (MPNet)  
**Backend:** FastAPI, REST APIs, WebSockets, SSE, Async Programming  
**Frontend:** React, Streamlit, Tailwind CSS  
**Databases:** SQLite (FTS5), Azure CosmosDB, Azure Blob Storage  
**Cloud & DevOps:** Microsoft Azure (ML, Cognitive Services, GPU), Docker  
**AI Systems:** Multi-Agent Systems, Function Calling, MCP (Model Context Protocol)  
**Tools:** Tesseract OCR, pdfplumber, NetworkX, PyVis, RapidFuzz, Joblib  
**Auth & Monitoring:** OAuth 2.0, MSAL, OpenTelemetry, Arize Phoenix

CERTIFICATIONS

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**Microsoft Azure Data Fundamentals (DP-900)** – Microsoft Azure | 2025  
**Prompt Engineering for Developers** – DeepLearning.AI (Andrew Ng) | 2024  
**Introduction to Large Language Models** – Google Cloud | 2024  
**Generative AI** – Google Cloud | 2024  
**Machine Learning** – NPTEL | 2024

LEADERSHIP & ACTIVITIES

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| <b>Event Lead</b>   | VIT Bhopal University |
| <i>Linpack Club</i> | <i>2023 – Present</i> |

- Organized and managed technical events and workshops, coordinating cross-functional teams and handling end-to-end event execution and participant engagement.

**Competitive Programming:** Solved 125+ Data Structures & Algorithms problems on LeetCode.  
**Sports:** Represented Thunders at Aavahan inter-college volleyball tournament, VIT Bhopal.